Hong Kong Institute of Information Technology

Higher Diploma in Software Engineering (IT114105)

Module: Contemporary Topics in Software Engineering (ITP4507)

Assignment One

Smart Delivery Fleet Management System (SDFMS)

Student name: _____

Student number: _____

Class: _____

1. Assumptions

1. The program is a single-user console application. All data is kept in memory and is discarded when the program exits. There is no database and no file.
2. The command codes are the lower-case codes in the sample. Any other text, including a capital letter, is an invalid command. Exit (`x`) stops the program and is not stored as a command.
3. Create fleet, add driver, modify driver role, delete driver and change fleet name can be undone and redone. Set current fleet, show fleet, display all fleets, list, undo and redo are not put on the undo list. This matches the sample undo list, which does not contain `s`, `sf`, `sa` or `l`.
4. A command that fails is not recorded. Examples are an unknown fleet id, an unknown driver, a duplicate id, an empty name and an invalid role.
5. Undo and redo are last-in, first-out. Doing a new undoable command clears the redo list.
6. Modify driver role and change fleet name use the Memento pattern for both undo and redo. Create, add and delete undo by the opposite action: remove the fleet, drop the driver, or put the driver back at the same position. The assignment asks for Memento on the two field changes, not on every command.
7. Each command remembers the fleet it changed. Undo still updates that fleet if the user has since selected another fleet.
8. Creating a fleet selects it as the current fleet. Undoing that create restores the previous current fleet when the removed fleet is still current.
9. Fleet ids are unique in the whole system. Driver ids are unique inside one fleet. The same driver id may be used in a different fleet.
10. A driver line is `id, name`. The first comma separates the two parts. Spaces around the id, the name, the fleet id and the fleet name are removed.

11. Role numbers belong to the fleet type. In a delivery van fleet, 1 is van driver and 2 is courier driver. In a refrigerated truck fleet, 1 is refrigerated-truck driver and 2 is cold-chain delivery driver. Any other number is rejected. The same number does not mean the same job in both fleets.
12. Show fleet always prints every role of that fleet. A role with no driver is printed as `Heading: NIL`. Drivers under a role stay in the order they were added. Display all fleets prints fleets in creation order and does not print drivers.
13. There is no current fleet until the first fleet is created. Add, modify, delete, show and change name then print `No current fleet!`.
14. The driver list is a `Vector` stored in an `AbstractList` field. The assignment allows Vector or LinkedList. The choice is made in one private method.
15. A new fleet type is added by writing a new `VehicleFleet` subclass and registering one factory type. The existing fleet classes and the command classes are not edited for that change.

2. User guide

On Windows, open a command prompt in the folder that contains the `.java` files and run `run.bat`. The batch file compiles every source file with `javac *.java` and starts `java Main`.

The menu is printed at the start of every action. After a fleet has been selected, the program also prints `The current fleet is <id> <name>.`

Code	Action	Further input
c	Create a fleet and select it	Type (`v` or `r`), fleet id, fleet name
s	Select the current fleet	Fleet id
a	Add a driver to the current fleet	`id, name` and a role number
m	Change a driver's role	Driver id and a role number
d	Delete a driver	Driver id
sf	Show the current fleet, including drivers	None
sa	List every fleet, without drivers	None
cn	Rename the current fleet	New name
u	Undo the last undoable command	None
r	Redo the last undone command	None
l	Show the undo list and the redo list	None
x	Exit	None

The role prompt depends on the current fleet. A delivery van asks for van driver or courier driver. A refrigerated truck asks for refrigerated-truck driver or cold-chain delivery driver. Undo and redo do not

ask again. They restore the saved result.

3. Application design

The given classes are kept. `VehicleFleet` is abstract. `updateDriverRole()` and `showFleet()` are implemented only in `DeliveryVanFleet` and `RefrigeratedTruckFleet`. `Driver` still has `driverID`, `driverName` and `role`, and the operations shown on the class diagram. The driver list is an `AbstractList`, and `getDrivers()` returns `Iterator<Driver>`.

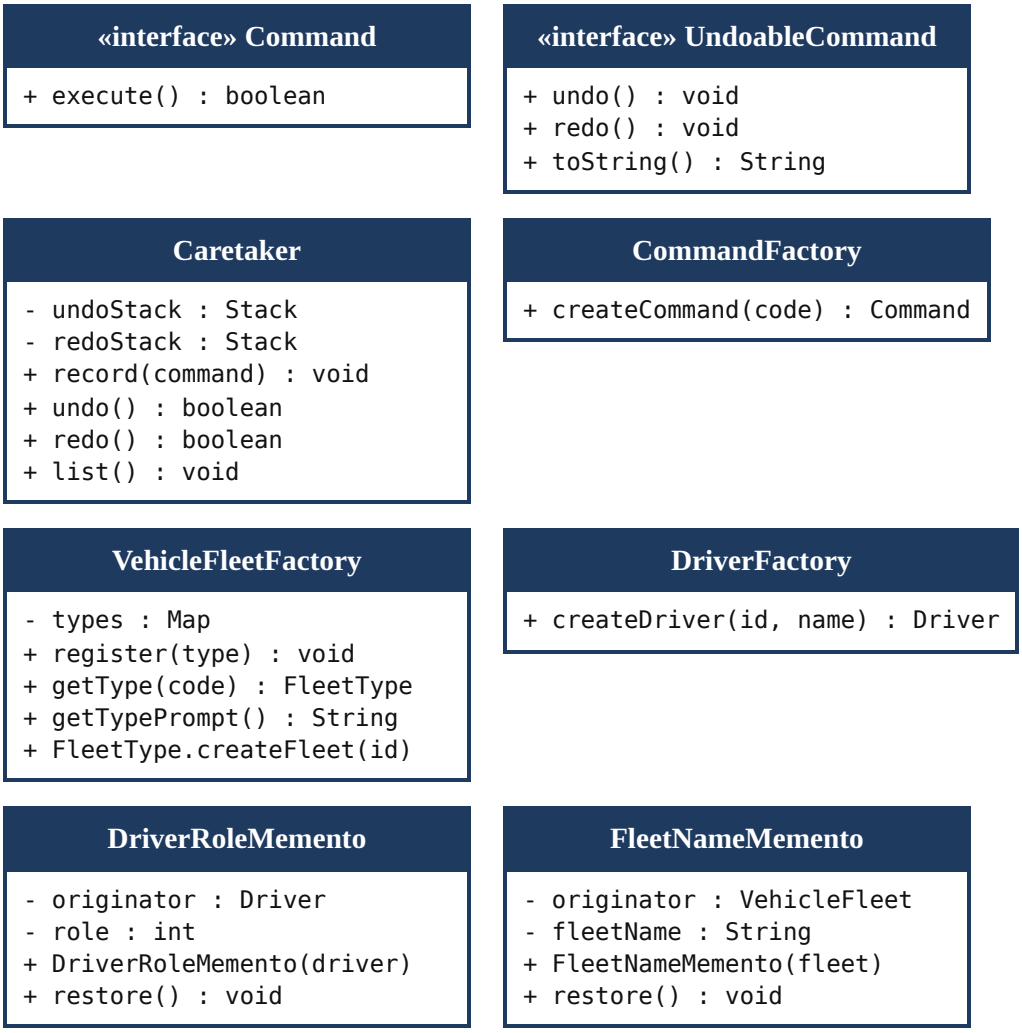
Extra operations are added only where the new behaviour needs them: finding a driver, putting a deleted driver back at the same index, and returning the display title. The fields of the given classes stay private.

3.1 Domain classes



`VehicleFleet` 1 has many `Driver` objects. `DeliveryVanFleet` and `RefrigeratedTruckFleet` inherit from `VehicleFleet`. `FleetSystem` holds the fleets in a `Vector` and keeps the current fleet.

3.2 Command, factory and memento



Concrete commands: CreateFleetCommand, SetCurrentFleetCommand, AddDriverCommand, ModifyDriverRoleCommand, DeleteDriverCommand, ShowFleetCommand, DisplayAllFleetsCommand, ChangeFleetNameCommand, UndoCommand, RedoCommand, ListUndoRedoCommand. The five state-changing commands implement UndoableCommand. The others implement Command only.

`Main` is the client. It prints the menu, asks the command factory for an object, calls `execute()`, and gives a successful undoable command to the caretaker. `FleetSystem`, `VehicleFleet` and `Driver` are the receivers.

4. Design patterns and lecture rules

4.1 Open-Closed Principle and embracing change

The lectures describe a module as open for extension and closed for modification. The part of SDFMS that is expected to grow is the fleet type. Role prompts, role names, the show-fleet layout and the class that is constructed all differ by type. Those differences are not written as an `if` on `"v"` or `"r"` inside the commands.

A new fleet, for example a motorcycle fleet, is a new subclass of `VehicleFleet` plus one `register` call on `VehicleFleetFactory`. `CreateFleetCommand`, `AddDriverCommand`, `ShowFleetCommand`

and `DisplayAllFleetsCommand` stay as they are. They call `updateDriverRole()`, `showFleet()`, `getRoleName()` and `getSummary()`. The subclass supplies the behaviour.

The lectures also say to separate code that may change. The concrete driver list is created in `VehicleFleet.createDriverList()`. Today it returns a `Vector`. Changing it to a `LinkedList` does not touch `addDriver`, `dropDriver` or `getDrivers`, because those use the `AbstractList` field. Role numbers are named constants in the subclass, not bare literals in the commands. Fields are private and are changed through methods, which is the encapsulation rule from the same notes. There is no global variable.

4.2 Factory

The factory lecture separates object creation from the code that uses the object. It shows a factory method such as `createList()` or `createCandyBar()`, and it shows an abstract factory whose concrete factories create one family of products. An `if` or `else` that decides which product to construct is the example given of a breach of the Open-Closed Principle, because a new product forces that method to be edited.

SDFMS uses three factories, not a factory class for every command:

- `VehicleFleetFactory` holds a `FleetType` for each fleet. `FleetType.createFleet(id)` is the factory method. `DeliveryVanType` returns a `DeliveryVanFleet`. `RefrigeratedTruckType` returns a `RefrigeratedTruckFleet`. `CreateFleetCommand` asks the factory for the type code and never writes `new DeliveryVanFleet`. Adding a type is a new `FleetType` and a `register` call.
- `DriverFactory.createDriver(id, name)` is the only place that constructs a `Driver`. It uses the diagram constructor and then `setName`.
- `CommandFactory.createCommand(code)` is the only place that constructs command objects. The menu codes are fixed by the assignment, so they live in this one class. A separate factory class for each of the eleven commands would repeat the same one-line constructor call and would not make a new fleet type easier to add.

The type prompt is built from the registered factories, so the sample text `Enter fleet type (v = delivery van | r = refrigerated truck)` is produced from the data rather than copied as a special case inside the command.

4.3 Command

`Command` has `execute()`. `execute()` returns false when the action cannot be completed, and that command is not recorded. `UndoableCommand` adds `undo()`, `redo()` and `toString()`. The history line is `toString()`, which is the printing method discussed in the polymorphism notes. The text is saved when the command runs, so a later rename does not rewrite an older "create fleet" line. The sample still shows `CENTRAL EXPRESS` on the create line after the fleet has been renamed.

`Main` is the client. `Caretaker` stores the undo stack and the redo stack. Undo pops the latest command, calls `undo()`, and pushes it onto the redo stack. Redo does the reverse. `record` clears the redo stack, so a new action cannot redo a branch that was thrown away. The stacks are `java.util.Stack`. Index 0 is the oldest entry, which is why the printed list matches the sample: the oldest command is at the top and the next undo is the last line.

Set current fleet, show, display all and list are commands, because the assignment asks for the Command pattern on those functions, but they do not implement `UndoableCommand`. Exit stays in `Main`. The required command list does not include exit.

4.4 Memento

The memento lecture uses this shape: the memento keeps the originator and a copy of the state, and `restore()` writes that state back.

```
public Memento(Originator o) {
    orig = o;
    state = orig.state;
}
public void restore() {
    orig.state = this.state;
}
```

`DriverRoleMemento` and `FleetNameMemento` follow that shape. The saved value is taken through `getRole()` or `getName()`, and `restore()` writes it back through `setRole()` or `setName()`. The lecture fragment assigns the field directly. This program uses the accessor instead, because the encapsulation rule says the field stays private. The result is the same: only the memento and the originator know how the old value is put back.

`ModifyDriverRoleCommand` creates a memento before `setRole` and another memento after it. Undo calls `beforeMemento.restore()`. Redo calls `afterMemento.restore()`. `ChangeFleetNameCommand` does the same with the fleet name. Neither undo nor redo asks the user again.

Create, add and delete are not field edits of one object. They add or remove an object from a collection. Their undo is the opposite action, and the command remembers the object and, for delete, the original index. Memento is not forced onto those commands. The assignment requires it for modify role and change name.

4.5 Polymorphism

`updateDriverRole()` and `showFleet()` are abstract. Each subclass overrides them with the same signature, as the polymorphism notes require. The van fleet prints the van role prompt and the van headings. The truck fleet prints the truck prompt and the truck headings. Shared printing of one role

group is in `printDriversForRole` on the superclass, so the two subclasses do not copy that loop. That is the "do not repeat yourself" point from the same lecture. No command asks `instanceof DeliveryVanFleet`.

4.6 Adapter

The Adapter lecture is for an existing class whose methods do not match the interface the client already calls. SDFMS does not have that problem. `DeliveryVanFleet` and `RefrigeratedTruckFleet` are written as subclasses of `VehicleFleet`, which is the interface the commands call. An adapter class would sit between two interfaces that this system does not have. It is not used.

5. Test plan and test cases

The tests were run with the compiled program. Input was typed into the real menu. The captures below are that output, not text written by hand. TC01 to TC15 are one continuous run and follow the sample. TC16 to TC25 are separate runs for the error and undo cases.

Pass means the messages, the fleet headings, the role groups and the undo text match the expected result for that case.

Case	Function	Result
TC01	Create delivery van fleet	Pass
TC02	Add van drivers	Pass
TC03	Show current fleet	Pass
TC04	Create refrigerated truck fleet and list all	Pass
TC05	Add refrigerated-truck drivers	Pass
TC06	Show refrigerated truck fleet	Pass
TC07	Set current fleet	Pass
TC08	Display all fleets	Pass
TC09	Modify driver role	Pass
TC10	Delete driver	Pass
TC11	Change fleet name	Pass
TC12	Show undo and redo lists	Pass
TC13	Undo	Pass
TC14	Redo	Pass
TC15	Errors and exit	Pass

TC16	No current fleet	Pass
TC17	Invalid create	Pass
TC18	Invalid driver input	Pass
TC19	Redo cleared by a new command	Pass
TC20	Delete undo keeps order	Pass
TC21	Undo hits the original fleet	Pass
TC22	Undo and redo of create	Pass
TC23	Empty new name	Pass
TC24	Failed commands are not recorded	Pass
TC25	Role memento on a truck fleet	Pass

TC01 Create a delivery van fleet

Steps. Command c, type v, id F001, name CENTRAL EXPRESS.

Expected result. The system prints "Delivery van fleet is created." and "Current fleet is changed to F001."

Result. Pass. The capture below is the program output.

```
TC01  Create a delivery van fleet

Smart Delivery Fleet Management System (SDFMS)
c = create fleet, s = set current fleet, a = add driver, m = modify driver role,
d = delete driver, sf = show fleet, sa = display all fleets, cn = change fleet name,
u = undo, r = redo, l = list undo/redo, x = exit system
Please enter command [ c | s | a | m | d | sf | sa | cn | u | r | l | x ] :- c
Enter fleet type (v = delivery van | r = refrigerated truck) :- v
Fleet ID:- F001
Fleet Name:- CENTRAL EXPRESS
Delivery van fleet is created.
Current fleet is changed to F001.
```

TC02 Add drivers to the delivery van fleet

Steps. Add d001 Alex Chan as role 1, then d002 Jamie Lee as role 2.

Expected result. Each add prints "Driver is added." The role prompt is the van-fleet prompt.

Result. Pass. The capture below is the program output.

```
TC02  Add drivers to the delivery van fleet

Smart Delivery Fleet Management System (SDFMS)
c = create fleet, s = set current fleet, a = add driver, m = modify driver role,
d = delete driver, sf = show fleet, sa = display all fleets, cn = change fleet name,
u = undo, r = redo, l = list undo/redo, x = exit system
The current fleet is F001 CENTRAL EXPRESS.
Please enter command [ c | s | a | m | d | sf | sa | cn | u | r | l | x ] :- a
Please input driver information (id, name):- d001, Alex Chan
Role (1 = van driver | 2 = courier driver) :- 1
Driver is added.

Smart Delivery Fleet Management System (SDFMS)
c = create fleet, s = set current fleet, a = add driver, m = modify driver role,
d = delete driver, sf = show fleet, sa = display all fleets, cn = change fleet name,
u = undo, r = redo, l = list undo/redo, x = exit system
The current fleet is F001 CENTRAL EXPRESS.
Please enter command [ c | s | a | m | d | sf | sa | cn | u | r | l | x ] :- a
Please input driver information (id, name):- d002, Jamie Lee
Role (1 = van driver | 2 = courier driver) :- 2
Driver is added.
```

TC03 Show the current fleet

Steps. Command sf while F001 is current.

Expected result. Van Driver lists d001. Courier Driver lists d002.

Result. Pass. The capture below is the program output.

```
TC03  Show the current fleet

Smart Delivery Fleet Management System (SDFMS)
c = create fleet, s = set current fleet, a = add driver, m = modify driver role,
d = delete driver, sf = show fleet, sa = display all fleets, cn = change fleet name,
u = undo, r = redo, l = list undo/redo, x = exit system
The current fleet is F001 CENTRAL EXPRESS.
Please enter command [ c | s | a | m | d | sf | sa | cn | u | r | l | x ] :- sf
Delivery Van Fleet CENTRAL EXPRESS (F001)
Van Driver:
d001, Alex Chan
Courier Driver:
d002, Jamie Lee
```

TC04 Create a refrigerated truck fleet and display all fleets

Steps. Command c, type r, id F102, name FROST LINE, then sa.

Expected result. The truck fleet is created and becomes current. sa lists F001 then F102, with no drivers.

Result. Pass. The capture below is the program output.

```
TC04  Create a refrigerated truck fleet and display all fleets

Smart Delivery Fleet Management System (SDFMS)
c = create fleet, s = set current fleet, a = add driver, m = modify driver role,
d = delete driver, sf = show fleet, sa = display all fleets, cn = change fleet name,
u = undo, r = redo, l = list undo/redo, x = exit system
The current fleet is F001 CENTRAL EXPRESS.
Please enter command [ c | s | a | m | d | sf | sa | cn | u | r | l | x ] :- c
Enter fleet type (v = delivery van | r = refrigerated truck) :- r
Fleet ID:- F102
Fleet Name:- FROST LINE
Refrigerated truck fleet is created.
Current fleet is changed to F102.

Smart Delivery Fleet Management System (SDFMS)
c = create fleet, s = set current fleet, a = add driver, m = modify driver role,
d = delete driver, sf = show fleet, sa = display all fleets, cn = change fleet name,
u = undo, r = redo, l = list undo/redo, x = exit system
The current fleet is F102 FROST LINE.
Please enter command [ c | s | a | m | d | sf | sa | cn | u | r | l | x ] :- sa
Delivery Van Fleet CENTRAL EXPRESS (F001)
Refrigerated Truck Fleet FROST LINE (F102)
```

TC05 Add drivers to the refrigerated truck fleet

Steps. Add d101 Morgan Wong as role 1, then d102 Taylor Lam as role 2.

Expected result. The role prompt is the refrigerated-truck prompt. Both drivers are added.

Result. Pass. The capture below is the program output.

```
TC05  Add drivers to the refrigerated truck fleet

Smart Delivery Fleet Management System (SDFMS)
c = create fleet, s = set current fleet, a = add driver, m = modify driver role,
d = delete driver, sf = show fleet, sa = display all fleets, cn = change fleet name,
u = undo, r = redo, l = list undo/redo, x = exit system
The current fleet is F102 FROST LINE.
Please enter command [ c | s | a | m | d | sf | sa | cn | u | r | l | x ] :- a
Please input driver information (id, name):- d101, Morgan Wong
Role (1 = refrigerated-truck driver | 2 = cold-chain delivery driver) :- 1
Driver is added.

Smart Delivery Fleet Management System (SDFMS)
c = create fleet, s = set current fleet, a = add driver, m = modify driver role,
d = delete driver, sf = show fleet, sa = display all fleets, cn = change fleet name,
u = undo, r = redo, l = list undo/redo, x = exit system
The current fleet is F102 FROST LINE.
Please enter command [ c | s | a | m | d | sf | sa | cn | u | r | l | x ] :- a
Please input driver information (id, name):- d102, Taylor Lam
Role (1 = refrigerated-truck driver | 2 = cold-chain delivery driver) :- 2
Driver is added.
```

TC06 Show the refrigerated truck fleet

Steps. Command sf while F102 is current.

Expected result. Refrigerated-Truck Driver lists d101. Cold-Chain Delivery Driver lists d102.

Result. Pass. The capture below is the program output.

```
TC06  Show the refrigerated truck fleet

Smart Delivery Fleet Management System (SDFMS)
c = create fleet, s = set current fleet, a = add driver, m = modify driver role,
d = delete driver, sf = show fleet, sa = display all fleets, cn = change fleet name,
u = undo, r = redo, l = list undo/redo, x = exit system
The current fleet is F102 FROST LINE.
Please enter command [ c | s | a | m | d | sf | sa | cn | u | r | l | x ] :- sf
Refrigerated Truck Fleet FROST LINE (F102)
Refrigerated-Truck Driver:
d101, Morgan Wong
Cold-Chain Delivery Driver:
d102, Taylor Lam
```

TC07 Set the current fleet by Fleet ID

Steps. Command s, fleet id F001.

Expected result. The system prints "Changed current fleet to F001."

Result. Pass. The capture below is the program output.

```
TC07  Set the current fleet by Fleet ID

Smart Delivery Fleet Management System (SDFMS)
c = create fleet, s = set current fleet, a = add driver, m = modify driver role,
d = delete driver, sf = show fleet, sa = display all fleets, cn = change fleet name,
u = undo, r = redo, l = list undo/redo, x = exit system
The current fleet is F102 FROST LINE.
Please enter command [ c | s | a | m | d | sf | sa | cn | u | r | l | x ] :- s
Please input fleet ID:- F001
Changed current fleet to F001.
```

TC08 Display all fleets

Steps. Command sa after switching back to F001.

Expected result. Both fleets are listed in creation order. Drivers are not listed.

Result. Pass. The capture below is the program output.

```
TC08  Display all fleets

Smart Delivery Fleet Management System (SDFMS)
c = create fleet, s = set current fleet, a = add driver, m = modify driver role,
d = delete driver, sf = show fleet, sa = display all fleets, cn = change fleet name,
u = undo, r = redo, l = list undo/redo, x = exit system
The current fleet is F001 CENTRAL EXPRESS.
Please enter command [ c | s | a | m | d | sf | sa | cn | u | r | l | x ] :- sa
Delivery Van Fleet CENTRAL EXPRESS (F001)
Refrigerated Truck Fleet FROST LINE (F102)
```

TC09 Modify a driver role

Steps. Command m, driver d001, role 2, then sf.

Expected result. d001 moves from Van Driver to Courier Driver. Van Driver is NIL.

Result. Pass. The capture below is the program output.

```
TC09  Modify a driver role

Smart Delivery Fleet Management System (SDFMS)
c = create fleet, s = set current fleet, a = add driver, m = modify driver role,
d = delete driver, sf = show fleet, sa = display all fleets, cn = change fleet name,
u = undo, r = redo, l = list undo/redo, x = exit system
The current fleet is F001 CENTRAL EXPRESS.
Please enter command [ c | s | a | m | d | sf | sa | cn | u | r | l | x ] :- m
Please input driver ID:- d001
Role (1 = van driver | 2 = courier driver) :- 2
Driver role is updated.

Smart Delivery Fleet Management System (SDFMS)
c = create fleet, s = set current fleet, a = add driver, m = modify driver role,
d = delete driver, sf = show fleet, sa = display all fleets, cn = change fleet name,
u = undo, r = redo, l = list undo/redo, x = exit system
The current fleet is F001 CENTRAL EXPRESS.
Please enter command [ c | s | a | m | d | sf | sa | cn | u | r | l | x ] :- sf
Delivery Van Fleet CENTRAL EXPRESS (F001)
Van Driver: NIL
Courier Driver:
d001, Alex Chan
d002, Jamie Lee
```

TC10 Delete a driver

Steps. Command d, driver d002, then sf.

Expected result. d002 is removed. Courier Driver lists only d001.

Result. Pass. The capture below is the program output.

```
TC10  Delete a driver

Smart Delivery Fleet Management System (SDFMS)
c = create fleet, s = set current fleet, a = add driver, m = modify driver role,
d = delete driver, sf = show fleet, sa = display all fleets, cn = change fleet name,
u = undo, r = redo, l = list undo/redo, x = exit system
The current fleet is F001 CENTRAL EXPRESS.
Please enter command [ c | s | a | m | d | sf | sa | cn | u | r | l | x ] :- d
Please input driver ID:- d002
Driver is deleted.

Smart Delivery Fleet Management System (SDFMS)
c = create fleet, s = set current fleet, a = add driver, m = modify driver role,
d = delete driver, sf = show fleet, sa = display all fleets, cn = change fleet name,
u = undo, r = redo, l = list undo/redo, x = exit system
The current fleet is F001 CENTRAL EXPRESS.
Please enter command [ c | s | a | m | d | sf | sa | cn | u | r | l | x ] :- sf
Delivery Van Fleet CENTRAL EXPRESS (F001)
Van Driver: NIL
Courier Driver:
d001, Alex Chan
```

TC11 Change the name of the current fleet

Steps. Command cn, new name METRO EXPRESS, then sf.

Expected result. The current fleet line and the show-fleet heading both use METRO EXPRESS.

Result. Pass. The capture below is the program output.

```
TC11  Change the name of the current fleet

Smart Delivery Fleet Management System (SDFMS)
c = create fleet, s = set current fleet, a = add driver, m = modify driver role,
d = delete driver, sf = show fleet, sa = display all fleets, cn = change fleet name,
u = undo, r = redo, l = list undo/redo, x = exit system
The current fleet is F001 CENTRAL EXPRESS.
Please enter command [ c | s | a | m | d | sf | sa | cn | u | r | l | x ] :- cn
Please input new name of the current fleet:- METRO EXPRESS
Fleet name is updated.

Smart Delivery Fleet Management System (SDFMS)
c = create fleet, s = set current fleet, a = add driver, m = modify driver role,
d = delete driver, sf = show fleet, sa = display all fleets, cn = change fleet name,
u = undo, r = redo, l = list undo/redo, x = exit system
The current fleet is F001 METRO EXPRESS.
Please enter command [ c | s | a | m | d | sf | sa | cn | u | r | l | x ] :- sf
Delivery Van Fleet METRO EXPRESS (F001)
Van Driver: NIL
Courier Driver:
d001, Alex Chan
```

TC12 Show undo/redo lists

Steps. Command l after the sample actions above.

Expected result. The undo list matches the sample, oldest first. The redo list is empty.

Result. Pass. The capture below is the program output.

```
TC12  Show undo/redo lists

Smart Delivery Fleet Management System (SDFMS)
c = create fleet, s = set current fleet, a = add driver, m = modify driver role,
d = delete driver, sf = show fleet, sa = display all fleets, cn = change fleet name,
u = undo, r = redo, l = list undo/redo, x = exit system
The current fleet is F001 METRO EXPRESS.
Please enter command [ c | s | a | m | d | sf | sa | cn | u | r | l | x ] :- l
Undo List
Create delivery van fleet, F001, CENTRAL EXPRESS
Add driver, d001, Alex Chan, van driver
Add driver, d002, Jamie Lee, courier driver
Create refrigerated truck fleet, F102, FROST LINE
Add driver, d101, Morgan Wong, refrigerated-truck driver
Add driver, d102, Taylor Lam, cold-chain delivery driver
Modify driver role, d001, courier driver
Delete driver, d002
Change fleet name, F001, METRO EXPRESS
-- End of undo list --
Redo List
-- End of redo list --
```

TC13 Undo

Steps. Command u three times, then sf.

Expected result. Name, delete and role change are undone in that order. F001 shows both original drivers and the original roles.

Result. Pass. The capture below is the program output.

```
TC13  Undo actions

Smart Delivery Fleet Management System (SDFMS)
c = create fleet, s = set current fleet, a = add driver, m = modify driver role,
d = delete driver, sf = show fleet, sa = display all fleets, cn = change fleet name,
u = undo, r = redo, l = list undo/redo, x = exit system
The current fleet is F001 METRO EXPRESS.
Please enter command [ c | s | a | m | d | sf | sa | cn | u | r | l | x ] :- u
Command (Change fleet name, F001, METRO EXPRESS) is undone.

Smart Delivery Fleet Management System (SDFMS)
c = create fleet, s = set current fleet, a = add driver, m = modify driver role,
d = delete driver, sf = show fleet, sa = display all fleets, cn = change fleet name,
u = undo, r = redo, l = list undo/redo, x = exit system
The current fleet is F001 CENTRAL EXPRESS.
Please enter command [ c | s | a | m | d | sf | sa | cn | u | r | l | x ] :- u
Command (Delete driver, d002) is undone.

Smart Delivery Fleet Management System (SDFMS)
c = create fleet, s = set current fleet, a = add driver, m = modify driver role,
d = delete driver, sf = show fleet, sa = display all fleets, cn = change fleet name,
u = undo, r = redo, l = list undo/redo, x = exit system
The current fleet is F001 CENTRAL EXPRESS.
Please enter command [ c | s | a | m | d | sf | sa | cn | u | r | l | x ] :- u
Command (Modify driver role, d001, courier driver) is undone.

Smart Delivery Fleet Management System (SDFMS)
c = create fleet, s = set current fleet, a = add driver, m = modify driver role,
d = delete driver, sf = show fleet, sa = display all fleets, cn = change fleet name,
u = undo, r = redo, l = list undo/redo, x = exit system
The current fleet is F001 CENTRAL EXPRESS.
Please enter command [ c | s | a | m | d | sf | sa | cn | u | r | l | x ] :- sf
Delivery Van Fleet CENTRAL EXPRESS (F001)
Van Driver:
d001, Alex Chan
Courier Driver:
d002, Jamie Lee
```

TC14 Redo

Steps. Command r three times, then sf.

Expected result. Role change, delete and name change are redone. The fleet is METRO EXPRESS again and d002 is gone.

Result. Pass. The capture below is the program output.

TC14 Redo actions

```
Smart Delivery Fleet Management System (SDFMS)
c = create fleet, s = set current fleet, a = add driver, m = modify driver role,
d = delete driver, sf = show fleet, sa = display all fleets, cn = change fleet name,
u = undo, r = redo, l = list undo/redo, x = exit system
The current fleet is F001 CENTRAL EXPRESS.
Please enter command [ c | s | a | m | d | sf | sa | cn | u | r | l | x ] :- r
Command (Modify driver role, d001, courier driver) is redone.

Smart Delivery Fleet Management System (SDFMS)
c = create fleet, s = set current fleet, a = add driver, m = modify driver role,
d = delete driver, sf = show fleet, sa = display all fleets, cn = change fleet name,
u = undo, r = redo, l = list undo/redo, x = exit system
The current fleet is F001 CENTRAL EXPRESS.
Please enter command [ c | s | a | m | d | sf | sa | cn | u | r | l | x ] :- r
Command (Delete driver, d002) is redone.

Smart Delivery Fleet Management System (SDFMS)
c = create fleet, s = set current fleet, a = add driver, m = modify driver role,
d = delete driver, sf = show fleet, sa = display all fleets, cn = change fleet name,
u = undo, r = redo, l = list undo/redo, x = exit system
The current fleet is F001 CENTRAL EXPRESS.
Please enter command [ c | s | a | m | d | sf | sa | cn | u | r | l | x ] :- r
Command (Change fleet name, F001, METRO EXPRESS) is redone.

Smart Delivery Fleet Management System (SDFMS)
c = create fleet, s = set current fleet, a = add driver, m = modify driver role,
d = delete driver, sf = show fleet, sa = display all fleets, cn = change fleet name,
u = undo, r = redo, l = list undo/redo, x = exit system
The current fleet is F001 METRO EXPRESS.
Please enter command [ c | s | a | m | d | sf | sa | cn | u | r | l | x ] :- sf
Delivery Van Fleet METRO EXPRESS (F001)
Van Driver: NIL
Courier Driver:
d001, Alex Chan
```

TC15 Unknown fleet, unknown driver, invalid command, exit

Steps. s F999, d d999, q, then x.

Expected result. The three error lines match the sample, then the system prints the termination line.

Result. Pass. The capture below is the program output.

```
TC15  Error handling and exit

Smart Delivery Fleet Management System (SDFMS)
c = create fleet, s = set current fleet, a = add driver, m = modify driver role,
d = delete driver, sf = show fleet, sa = display all fleets, cn = change fleet name,
u = undo, r = redo, l = list undo/redo, x = exit system
The current fleet is F001 METRO EXPRESS.
Please enter command [ c | s | a | m | d | sf | sa | cn | u | r | l | x ] :- s
Please input fleet ID:- F999
Fleet F999 is not found!

Smart Delivery Fleet Management System (SDFMS)
c = create fleet, s = set current fleet, a = add driver, m = modify driver role,
d = delete driver, sf = show fleet, sa = display all fleets, cn = change fleet name,
u = undo, r = redo, l = list undo/redo, x = exit system
The current fleet is F001 METRO EXPRESS.
Please enter command [ c | s | a | m | d | sf | sa | cn | u | r | l | x ] :- d
Please input driver ID:- d999
Driver d999 is not found in the current fleet!

Smart Delivery Fleet Management System (SDFMS)
c = create fleet, s = set current fleet, a = add driver, m = modify driver role,
d = delete driver, sf = show fleet, sa = display all fleets, cn = change fleet name,
u = undo, r = redo, l = list undo/redo, x = exit system
The current fleet is F001 METRO EXPRESS.
Please enter command [ c | s | a | m | d | sf | sa | cn | u | r | l | x ] :- q
Invalid command!

Smart Delivery Fleet Management System (SDFMS)
c = create fleet, s = set current fleet, a = add driver, m = modify driver role,
d = delete driver, sf = show fleet, sa = display all fleets, cn = change fleet name,
u = undo, r = redo, l = list undo/redo, x = exit system
The current fleet is F001 METRO EXPRESS.
Please enter command [ c | s | a | m | d | sf | sa | cn | u | r | l | x ] :- x
Thank you for using SDFMS. System terminated.
```

TC16 No fleet has been created yet

Steps. a, m, d, sf, cn, sa, u, r, then x, on a new run.

Expected result. Add, modify, delete, show and rename report that there is no current fleet. sa reports no fleet record. Undo and redo report empty lists.

Result. Pass. The capture below is the program output.

TC16 Commands before any fleet exists (1/2)

```
Smart Delivery Fleet Management System (SDFMS)
c = create fleet, s = set current fleet, a = add driver, m = modify driver role,
d = delete driver, sf = show fleet, sa = display all fleets, cn = change fleet name,
u = undo, r = redo, l = list undo/redo, x = exit system
Please enter command [ c | s | a | m | d | sf | sa | cn | u | r | l | x ] :- a
No current fleet!
Smart Delivery Fleet Management System (SDFMS)
c = create fleet, s = set current fleet, a = add driver, m = modify driver role,
d = delete driver, sf = show fleet, sa = display all fleets, cn = change fleet name,
u = undo, r = redo, l = list undo/redo, x = exit system
Please enter command [ c | s | a | m | d | sf | sa | cn | u | r | l | x ] :- m
No current fleet!
Smart Delivery Fleet Management System (SDFMS)
c = create fleet, s = set current fleet, a = add driver, m = modify driver role,
d = delete driver, sf = show fleet, sa = display all fleets, cn = change fleet name,
u = undo, r = redo, l = list undo/redo, x = exit system
Please enter command [ c | s | a | m | d | sf | sa | cn | u | r | l | x ] :- d
No current fleet!
Smart Delivery Fleet Management System (SDFMS)
c = create fleet, s = set current fleet, a = add driver, m = modify driver role,
d = delete driver, sf = show fleet, sa = display all fleets, cn = change fleet name,
u = undo, r = redo, l = list undo/redo, x = exit system
Please enter command [ c | s | a | m | d | sf | sa | cn | u | r | l | x ] :- sf
No current fleet!
Smart Delivery Fleet Management System (SDFMS)
c = create fleet, s = set current fleet, a = add driver, m = modify driver role,
d = delete driver, sf = show fleet, sa = display all fleets, cn = change fleet name,
u = undo, r = redo, l = list undo/redo, x = exit system
Please enter command [ c | s | a | m | d | sf | sa | cn | u | r | l | x ] :- cn
No current fleet!
Smart Delivery Fleet Management System (SDFMS)
c = create fleet, s = set current fleet, a = add driver, m = modify driver role,
d = delete driver, sf = show fleet, sa = display all fleets, cn = change fleet name,
u = undo, r = redo, l = list undo/redo, x = exit system
Please enter command [ c | s | a | m | d | sf | sa | cn | u | r | l | x ] :- sa
No fleet record!
Smart Delivery Fleet Management System (SDFMS)
c = create fleet, s = set current fleet, a = add driver, m = modify driver role,
d = delete driver, sf = show fleet, sa = display all fleets, cn = change fleet name,
u = undo, r = redo, l = list undo/redo, x = exit system
```

TC16 Commands before any fleet exists (2/2)

```
Please enter command [ c | s | a | m | d | sf | sa | cn | u | r | l | x ] :- u
Nothing to undo!
Smart Delivery Fleet Management System (SDFMS)
c = create fleet, s = set current fleet, a = add driver, m = modify driver role,
d = delete driver, sf = show fleet, sa = display all fleets, cn = change fleet name,
u = undo, r = redo, l = list undo/redo, x = exit system
Please enter command [ c | s | a | m | d | sf | sa | cn | u | r | l | x ] :- r
Nothing to redo!
Smart Delivery Fleet Management System (SDFMS)
c = create fleet, s = set current fleet, a = add driver, m = modify driver role,
d = delete driver, sf = show fleet, sa = display all fleets, cn = change fleet name,
u = undo, r = redo, l = list undo/redo, x = exit system
Please enter command [ c | s | a | m | d | sf | sa | cn | u | r | l | x ] :- x
Thank you for using SDFMS. System terminated.
```

TC17 Invalid fleet data

Steps. Unknown type, empty id, empty name, then a duplicate id.

Expected result. Nothing is created for the bad input. The duplicate id is rejected. Only the successful create is on the undo list.

Result. Pass. The capture below is the program output.

TC17 Invalid type, empty fields, duplicate fleet id (1/2)

```
Smart Delivery Fleet Management System (SDFMS)
c = create fleet, s = set current fleet, a = add driver, m = modify driver role,
d = delete driver, sf = show fleet, sa = display all fleets, cn = change fleet name,
u = undo, r = redo, l = list undo/redo, x = exit system
Please enter command [ c | s | a | m | d | sf | sa | cn | u | r | l | x ] :- c
Enter fleet type (v = delivery van | r = refrigerated truck) :- bike
Invalid fleet type!
Smart Delivery Fleet Management System (SDFMS)
c = create fleet, s = set current fleet, a = add driver, m = modify driver role,
d = delete driver, sf = show fleet, sa = display all fleets, cn = change fleet name,
u = undo, r = redo, l = list undo/redo, x = exit system
Please enter command [ c | s | a | m | d | sf | sa | cn | u | r | l | x ] :- c
Enter fleet type (v = delivery van | r = refrigerated truck) :- v
Fleet ID:-
Fleet ID cannot be empty!
Smart Delivery Fleet Management System (SDFMS)
c = create fleet, s = set current fleet, a = add driver, m = modify driver role,
d = delete driver, sf = show fleet, sa = display all fleets, cn = change fleet name,
u = undo, r = redo, l = list undo/redo, x = exit system
Please enter command [ c | s | a | m | d | sf | sa | cn | u | r | l | x ] :- c
Enter fleet type (v = delivery van | r = refrigerated truck) :- v
Fleet ID:- F001
Fleet Name:-
Fleet name cannot be empty!
Smart Delivery Fleet Management System (SDFMS)
c = create fleet, s = set current fleet, a = add driver, m = modify driver role,
d = delete driver, sf = show fleet, sa = display all fleets, cn = change fleet name,
u = undo, r = redo, l = list undo/redo, x = exit system
Please enter command [ c | s | a | m | d | sf | sa | cn | u | r | l | x ] :- c
Enter fleet type (v = delivery van | r = refrigerated truck) :- v
Fleet ID:- F001
Fleet Name:- ALPHA
Delivery van fleet is created.
Current fleet is changed to F001.
Smart Delivery Fleet Management System (SDFMS)
c = create fleet, s = set current fleet, a = add driver, m = modify driver role,
d = delete driver, sf = show fleet, sa = display all fleets, cn = change fleet name,
u = undo, r = redo, l = list undo/redo, x = exit system
The current fleet is F001 ALPHA.
Please enter command [ c | s | a | m | d | sf | sa | cn | u | r | l | x ] :- c
```

TC17 Invalid type, empty fields, duplicate fleet id (2/2)

```
Enter fleet type (v = delivery van | r = refrigerated truck) :- v
Fleet ID:- F001
Fleet ID already exists!
Smart Delivery Fleet Management System (SDFMS)
c = create fleet, s = set current fleet, a = add driver, m = modify driver role,
d = delete driver, sf = show fleet, sa = display all fleets, cn = change fleet name,
u = undo, r = redo, l = list undo/redo, x = exit system
The current fleet is F001 ALPHA.
Please enter command [ c | s | a | m | d | sf | sa | cn | u | r | l | x ] :- OTHER
Invalid command!
Smart Delivery Fleet Management System (SDFMS)
c = create fleet, s = set current fleet, a = add driver, m = modify driver role,
d = delete driver, sf = show fleet, sa = display all fleets, cn = change fleet name,
u = undo, r = redo, l = list undo/redo, x = exit system
The current fleet is F001 ALPHA.
Please enter command [ c | s | a | m | d | sf | sa | cn | u | r | l | x ] :- l
Undo List
Create delivery van fleet, F001, ALPHA
-- End of undo list --
Redo List
-- End of redo list --
Smart Delivery Fleet Management System (SDFMS)
c = create fleet, s = set current fleet, a = add driver, m = modify driver role,
d = delete driver, sf = show fleet, sa = display all fleets, cn = change fleet name,
u = undo, r = redo, l = list undo/redo, x = exit system
The current fleet is F001 ALPHA.
Please enter command [ c | s | a | m | d | sf | sa | cn | u | r | l | x ] :- x
Thank you for using SDFMS. System terminated.
```

TC18 Invalid driver data

Steps. Bad id/name text, illegal role, duplicate id, unknown driver, then a valid role change.

Expected result. Only a valid add and a valid role change are stored. The show result has the driver under Courier Driver.

Result. Pass. The capture below is the program output.

```
TC18  Invalid driver input, duplicate driver, invalid role  (1/3)

Smart Delivery Fleet Management System (SDFMS)
c = create fleet, s = set current fleet, a = add driver, m = modify driver role,
d = delete driver, sf = show fleet, sa = display all fleets, cn = change fleet name,
u = undo, r = redo, l = list undo/redo, x = exit system
Please enter command [ c | s | a | m | d | sf | sa | cn | u | r | l | x ] :- c
Enter fleet type (v = delivery van | r = refrigerated truck) :- v
Fleet ID:- F010
Fleet Name:- CITY
Delivery van fleet is created.
Current fleet is changed to F010.
Smart Delivery Fleet Management System (SDFMS)
c = create fleet, s = set current fleet, a = add driver, m = modify driver role,
d = delete driver, sf = show fleet, sa = display all fleets, cn = change fleet name,
u = undo, r = redo, l = list undo/redo, x = exit system
The current fleet is F010 CITY.
Please enter command [ c | s | a | m | d | sf | sa | cn | u | r | l | x ] :- a
Please input driver information (id, name):- just-id
Invalid driver information!
Smart Delivery Fleet Management System (SDFMS)
c = create fleet, s = set current fleet, a = add driver, m = modify driver role,
d = delete driver, sf = show fleet, sa = display all fleets, cn = change fleet name,
u = undo, r = redo, l = list undo/redo, x = exit system
The current fleet is F010 CITY.
Please enter command [ c | s | a | m | d | sf | sa | cn | u | r | l | x ] :- a
Please input driver information (id, name):- , Sam
Invalid driver information!
Smart Delivery Fleet Management System (SDFMS)
c = create fleet, s = set current fleet, a = add driver, m = modify driver role,
d = delete driver, sf = show fleet, sa = display all fleets, cn = change fleet name,
u = undo, r = redo, l = list undo/redo, x = exit system
The current fleet is F010 CITY.
Please enter command [ c | s | a | m | d | sf | sa | cn | u | r | l | x ] :- a
Please input driver information (id, name):- d1,
Invalid driver information!
Smart Delivery Fleet Management System (SDFMS)
c = create fleet, s = set current fleet, a = add driver, m = modify driver role,
d = delete driver, sf = show fleet, sa = display all fleets, cn = change fleet name,
u = undo, r = redo, l = list undo/redo, x = exit system
The current fleet is F010 CITY.
Please enter command [ c | s | a | m | d | sf | sa | cn | u | r | l | x ] :- a
```

TC18 Invalid driver input, duplicate driver, invalid role (2/3)

```
Please input driver information (id, name):- d1, Sam
Role (1 = van driver | 2 = courier driver) :- 0
Invalid role!
Smart Delivery Fleet Management System (SDFMS)
c = create fleet, s = set current fleet, a = add driver, m = modify driver role,
d = delete driver, sf = show fleet, sa = display all fleets, cn = change fleet name,
u = undo, r = redo, l = list undo/redo, x = exit system
The current fleet is F010 CITY.
Please enter command [ c | s | a | m | d | sf | sa | cn | u | r | l | x ] :- a
Please input driver information (id, name):- d1, Sam
Role (1 = van driver | 2 = courier driver) :- abc
Invalid role!
Smart Delivery Fleet Management System (SDFMS)
c = create fleet, s = set current fleet, a = add driver, m = modify driver role,
d = delete driver, sf = show fleet, sa = display all fleets, cn = change fleet name,
u = undo, r = redo, l = list undo/redo, x = exit system
The current fleet is F010 CITY.
Please enter command [ c | s | a | m | d | sf | sa | cn | u | r | l | x ] :- a
Please input driver information (id, name):- d1, Sam
Role (1 = van driver | 2 = courier driver) :- 1
Driver is added.
Smart Delivery Fleet Management System (SDFMS)
c = create fleet, s = set current fleet, a = add driver, m = modify driver role,
d = delete driver, sf = show fleet, sa = display all fleets, cn = change fleet name,
u = undo, r = redo, l = list undo/redo, x = exit system
The current fleet is F010 CITY.
Please enter command [ c | s | a | m | d | sf | sa | cn | u | r | l | x ] :- a
Please input driver information (id, name):- d1, Sam
Driver d1 already exists in the current fleet!
Smart Delivery Fleet Management System (SDFMS)
c = create fleet, s = set current fleet, a = add driver, m = modify driver role,
d = delete driver, sf = show fleet, sa = display all fleets, cn = change fleet name,
u = undo, r = redo, l = list undo/redo, x = exit system
The current fleet is F010 CITY.
Please enter command [ c | s | a | m | d | sf | sa | cn | u | r | l | x ] :- 2
Invalid command!
Smart Delivery Fleet Management System (SDFMS)
c = create fleet, s = set current fleet, a = add driver, m = modify driver role,
d = delete driver, sf = show fleet, sa = display all fleets, cn = change fleet name,
u = undo, r = redo, l = list undo/redo, x = exit system
```

```
TC18  Invalid driver input, duplicate driver, invalid role  (3/3)

The current fleet is F010 CITY.
Please enter command [ c | s | a | m | d | sf | sa | cn | u | r | l | x ] :- m
Please input driver ID:- missing
Driver missing is not found in the current fleet!
Smart Delivery Fleet Management System (SDFMS)
c = create fleet, s = set current fleet, a = add driver, m = modify driver role,
d = delete driver, sf = show fleet, sa = display all fleets, cn = change fleet name,
u = undo, r = redo, l = list undo/redo, x = exit system
The current fleet is F010 CITY.
Please enter command [ c | s | a | m | d | sf | sa | cn | u | r | l | x ] :- m
Please input driver ID:- d1
Role (1 = van driver | 2 = courier driver) :- 9
Invalid role!
Smart Delivery Fleet Management System (SDFMS)
c = create fleet, s = set current fleet, a = add driver, m = modify driver role,
d = delete driver, sf = show fleet, sa = display all fleets, cn = change fleet name,
u = undo, r = redo, l = list undo/redo, x = exit system
The current fleet is F010 CITY.
Please enter command [ c | s | a | m | d | sf | sa | cn | u | r | l | x ] :- m
Please input driver ID:- d1
Role (1 = van driver | 2 = courier driver) :- 2
Driver role is updated.
Smart Delivery Fleet Management System (SDFMS)
c = create fleet, s = set current fleet, a = add driver, m = modify driver role,
d = delete driver, sf = show fleet, sa = display all fleets, cn = change fleet name,
u = undo, r = redo, l = list undo/redo, x = exit system
The current fleet is F010 CITY.
Please enter command [ c | s | a | m | d | sf | sa | cn | u | r | l | x ] :- sf
Delivery Van Fleet CITY (F010)
Van Driver: NIL
Courier Driver:
d1, Sam
Smart Delivery Fleet Management System (SDFMS)
c = create fleet, s = set current fleet, a = add driver, m = modify driver role,
d = delete driver, sf = show fleet, sa = display all fleets, cn = change fleet name,
u = undo, r = redo, l = list undo/redo, x = exit system
The current fleet is F010 CITY.
Please enter command [ c | s | a | m | d | sf | sa | cn | u | r | l | x ] :- x
Thank you for using SDFMS. System terminated.
```

TC19 A new command clears redo

Steps. Undo a rename, add a driver, then try to redo.

Expected result. Redo says there is nothing to redo. The undo list contains the create and the add, not the undone rename.

Result. Pass. The capture below is the program output.

TC19 Empty history, and a new command clears redo (1/2)

```
Smart Delivery Fleet Management System (SDFMS)
c = create fleet, s = set current fleet, a = add driver, m = modify driver role,
d = delete driver, sf = show fleet, sa = display all fleets, cn = change fleet name,
u = undo, r = redo, l = list undo/redo, x = exit system
Please enter command [ c | s | a | m | d | sf | sa | cn | u | r | l | x ] :- u
Nothing to undo!
Smart Delivery Fleet Management System (SDFMS)
c = create fleet, s = set current fleet, a = add driver, m = modify driver role,
d = delete driver, sf = show fleet, sa = display all fleets, cn = change fleet name,
u = undo, r = redo, l = list undo/redo, x = exit system
Please enter command [ c | s | a | m | d | sf | sa | cn | u | r | l | x ] :- r
Nothing to redo!
Smart Delivery Fleet Management System (SDFMS)
c = create fleet, s = set current fleet, a = add driver, m = modify driver role,
d = delete driver, sf = show fleet, sa = display all fleets, cn = change fleet name,
u = undo, r = redo, l = list undo/redo, x = exit system
Please enter command [ c | s | a | m | d | sf | sa | cn | u | r | l | x ] :- c
Enter fleet type (v = delivery van | r = refrigerated truck) :- v
Fleet ID:- F001
Fleet Name:- ALPHA
Delivery van fleet is created.
Current fleet is changed to F001.
Smart Delivery Fleet Management System (SDFMS)
c = create fleet, s = set current fleet, a = add driver, m = modify driver role,
d = delete driver, sf = show fleet, sa = display all fleets, cn = change fleet name,
u = undo, r = redo, l = list undo/redo, x = exit system
The current fleet is F001 ALPHA.
Please enter command [ c | s | a | m | d | sf | sa | cn | u | r | l | x ] :- cn
Please input new name of the current fleet:- BETA
Fleet name is updated.
Smart Delivery Fleet Management System (SDFMS)
c = create fleet, s = set current fleet, a = add driver, m = modify driver role,
d = delete driver, sf = show fleet, sa = display all fleets, cn = change fleet name,
u = undo, r = redo, l = list undo/redo, x = exit system
The current fleet is F001 BETA.
Please enter command [ c | s | a | m | d | sf | sa | cn | u | r | l | x ] :- u
Command (Change fleet name, F001, BETA) is undone.
Smart Delivery Fleet Management System (SDFMS)
c = create fleet, s = set current fleet, a = add driver, m = modify driver role,
d = delete driver, sf = show fleet, sa = display all fleets, cn = change fleet name,
```

```
TC19  Empty history, and a new command clears redo  (2/2)

u = undo, r = redo, l = list undo/redo, x = exit system
The current fleet is F001 ALPHA.
Please enter command [ c | s | a | m | d | sf | sa | cn | u | r | l | x ] :- a
Please input driver information (id, name):- d1, Ann
Role (1 = van driver | 2 = courier driver) :- 1
Driver is added.
Smart Delivery Fleet Management System (SDFMS)
c = create fleet, s = set current fleet, a = add driver, m = modify driver role,
d = delete driver, sf = show fleet, sa = display all fleets, cn = change fleet name,
u = undo, r = redo, l = list undo/redo, x = exit system
The current fleet is F001 ALPHA.
Please enter command [ c | s | a | m | d | sf | sa | cn | u | r | l | x ] :- r
Nothing to redo!
Smart Delivery Fleet Management System (SDFMS)
c = create fleet, s = set current fleet, a = add driver, m = modify driver role,
d = delete driver, sf = show fleet, sa = display all fleets, cn = change fleet name,
u = undo, r = redo, l = list undo/redo, x = exit system
The current fleet is F001 ALPHA.
Please enter command [ c | s | a | m | d | sf | sa | cn | u | r | l | x ] :- l
Undo List
Create delivery van fleet, F001, ALPHA
Add driver, d1, Ann, van driver
-- End of undo list --
Redo List
-- End of redo list --
Smart Delivery Fleet Management System (SDFMS)
c = create fleet, s = set current fleet, a = add driver, m = modify driver role,
d = delete driver, sf = show fleet, sa = display all fleets, cn = change fleet name,
u = undo, r = redo, l = list undo/redo, x = exit system
The current fleet is F001 ALPHA.
Please enter command [ c | s | a | m | d | sf | sa | cn | u | r | l | x ] :- x
Thank you for using SDFMS. System terminated.
```

TC20 Undo of delete keeps driver order

Steps. Add Ann, Ben and Cara. Delete Ben. Undo.

Expected result. After the delete, Cara is the only courier. After undo, Ben is listed before Cara again.

Result. Pass. The capture below is the program output.

TC20 Undo delete puts the driver back in the original order (1/2)

```
Smart Delivery Fleet Management System (SDFMS)
c = create fleet, s = set current fleet, a = add driver, m = modify driver role,
d = delete driver, sf = show fleet, sa = display all fleets, cn = change fleet name,
u = undo, r = redo, l = list undo/redo, x = exit system
Please enter command [ c | s | a | m | d | sf | sa | cn | u | r | l | x ] :- c
Enter fleet type (v = delivery van | r = refrigerated truck) :- v
Fleet ID:- F001
Fleet Name:- ALPHA
Delivery van fleet is created.
Current fleet is changed to F001.
Smart Delivery Fleet Management System (SDFMS)
c = create fleet, s = set current fleet, a = add driver, m = modify driver role,
d = delete driver, sf = show fleet, sa = display all fleets, cn = change fleet name,
u = undo, r = redo, l = list undo/redo, x = exit system
The current fleet is F001 ALPHA.
Please enter command [ c | s | a | m | d | sf | sa | cn | u | r | l | x ] :- a
Please input driver information (id, name):- d1, Ann
Role (1 = van driver | 2 = courier driver) :- 1
Driver is added.
Smart Delivery Fleet Management System (SDFMS)
c = create fleet, s = set current fleet, a = add driver, m = modify driver role,
d = delete driver, sf = show fleet, sa = display all fleets, cn = change fleet name,
u = undo, r = redo, l = list undo/redo, x = exit system
The current fleet is F001 ALPHA.
Please enter command [ c | s | a | m | d | sf | sa | cn | u | r | l | x ] :- a
Please input driver information (id, name):- d2, Ben
Role (1 = van driver | 2 = courier driver) :- 2
Driver is added.
Smart Delivery Fleet Management System (SDFMS)
c = create fleet, s = set current fleet, a = add driver, m = modify driver role,
d = delete driver, sf = show fleet, sa = display all fleets, cn = change fleet name,
u = undo, r = redo, l = list undo/redo, x = exit system
The current fleet is F001 ALPHA.
Please enter command [ c | s | a | m | d | sf | sa | cn | u | r | l | x ] :- a
Please input driver information (id, name):- d3, Cara
Role (1 = van driver | 2 = courier driver) :- 2
Driver is added.
Smart Delivery Fleet Management System (SDFMS)
c = create fleet, s = set current fleet, a = add driver, m = modify driver role,
d = delete driver, sf = show fleet, sa = display all fleets, cn = change fleet name,
```

TC20 Undo delete puts the driver back in the original order (2/2)

```
u = undo, r = redo, l = list undo/redo, x = exit system
The current fleet is F001 ALPHA.
Please enter command [ c | s | a | m | d | sf | sa | cn | u | r | l | x ] :- d
Please input driver ID:- d2
Driver is deleted.
Smart Delivery Fleet Management System (SDFMS)
c = create fleet, s = set current fleet, a = add driver, m = modify driver role,
d = delete driver, sf = show fleet, sa = display all fleets, cn = change fleet name,
u = undo, r = redo, l = list undo/redo, x = exit system
The current fleet is F001 ALPHA.
Please enter command [ c | s | a | m | d | sf | sa | cn | u | r | l | x ] :- sf
Delivery Van Fleet ALPHA (F001)
Van Driver:
d1, Ann
Courier Driver:
d3, Cara
Smart Delivery Fleet Management System (SDFMS)
c = create fleet, s = set current fleet, a = add driver, m = modify driver role,
d = delete driver, sf = show fleet, sa = display all fleets, cn = change fleet name,
u = undo, r = redo, l = list undo/redo, x = exit system
The current fleet is F001 ALPHA.
Please enter command [ c | s | a | m | d | sf | sa | cn | u | r | l | x ] :- u
Command (Delete driver, d2) is undone.
Smart Delivery Fleet Management System (SDFMS)
c = create fleet, s = set current fleet, a = add driver, m = modify driver role,
d = delete driver, sf = show fleet, sa = display all fleets, cn = change fleet name,
u = undo, r = redo, l = list undo/redo, x = exit system
The current fleet is F001 ALPHA.
Please enter command [ c | s | a | m | d | sf | sa | cn | u | r | l | x ] :- sf
Delivery Van Fleet ALPHA (F001)
Van Driver:
d1, Ann
Courier Driver:
d2, Ben
d3, Cara
Smart Delivery Fleet Management System (SDFMS)
c = create fleet, s = set current fleet, a = add driver, m = modify driver role,
d = delete driver, sf = show fleet, sa = display all fleets, cn = change fleet name,
u = undo, r = redo, l = list undo/redo, x = exit system
The current fleet is F001 ALPHA.
Please enter command [ c | s | a | m | d | sf | sa | cn | u | r | l | x ] :- x
Thank you for using SDFMS. System terminated.
```

TC20 Undo delete puts the driver back in the original order (3/3)

```
Please enter command [ c | s | a | m | d | sf | sa | cn | u | r | l | x ] :- x
Thank you for using SDFMS. System terminated.
```

TC21 Undo applies to the fleet the command used

Steps. Add a driver and rename F001, create F002, add a truck driver, switch to F001, then undo.

Expected result. Undoing the truck add does not remove Alex from F001. Further undos remove F002 and restore the original name of F001.

Result. Pass. The capture below is the program output.

TC21 Undo updates the fleet the command was run on (1/3)

```
Smart Delivery Fleet Management System (SDFMS)
c = create fleet, s = set current fleet, a = add driver, m = modify driver role,
d = delete driver, sf = show fleet, sa = display all fleets, cn = change fleet name,
u = undo, r = redo, l = list undo/redo, x = exit system
Please enter command [ c | s | a | m | d | sf | sa | cn | u | r | l | x ] :- c
Enter fleet type (v = delivery van | r = refrigerated truck) :- v
Fleet ID:- F001
Fleet Name:- ALPHA
Delivery van fleet is created.
Current fleet is changed to F001.
Smart Delivery Fleet Management System (SDFMS)
c = create fleet, s = set current fleet, a = add driver, m = modify driver role,
d = delete driver, sf = show fleet, sa = display all fleets, cn = change fleet name,
u = undo, r = redo, l = list undo/redo, x = exit system
The current fleet is F001 ALPHA.
Please enter command [ c | s | a | m | d | sf | sa | cn | u | r | l | x ] :- a
Please input driver information (id, name):- d001, Alex
Role (1 = van driver | 2 = courier driver) :- 1
Driver is added.
Smart Delivery Fleet Management System (SDFMS)
c = create fleet, s = set current fleet, a = add driver, m = modify driver role,
d = delete driver, sf = show fleet, sa = display all fleets, cn = change fleet name,
u = undo, r = redo, l = list undo/redo, x = exit system
The current fleet is F001 ALPHA.
Please enter command [ c | s | a | m | d | sf | sa | cn | u | r | l | x ] :- cn
Please input new name of the current fleet:- ALPHA TWO
Fleet name is updated.
Smart Delivery Fleet Management System (SDFMS)
c = create fleet, s = set current fleet, a = add driver, m = modify driver role,
d = delete driver, sf = show fleet, sa = display all fleets, cn = change fleet name,
u = undo, r = redo, l = list undo/redo, x = exit system
The current fleet is F001 ALPHA TWO.
Please enter command [ c | s | a | m | d | sf | sa | cn | u | r | l | x ] :- c
Enter fleet type (v = delivery van | r = refrigerated truck) :- r
Fleet ID:- F002
Fleet Name:- BETA
Refrigerated truck fleet is created.
Current fleet is changed to F002.
Smart Delivery Fleet Management System (SDFMS)
c = create fleet, s = set current fleet, a = add driver, m = modify driver role,
```

TC21 Undo updates the fleet the command was run on (2/3)

```
d = delete driver, sf = show fleet, sa = display all fleets, cn = change fleet name,
u = undo, r = redo, l = list undo/redo, x = exit system
The current fleet is F002 BETA.
Please enter command [ c | s | a | m | d | sf | sa | cn | u | r | l | x ] :- a
Please input driver information (id, name):- d101, Morgan
Role (1 = refrigerated-truck driver | 2 = cold-chain delivery driver) :- 1
Driver is added.
Smart Delivery Fleet Management System (SDFMS)
c = create fleet, s = set current fleet, a = add driver, m = modify driver role,
d = delete driver, sf = show fleet, sa = display all fleets, cn = change fleet name,
u = undo, r = redo, l = list undo/redo, x = exit system
The current fleet is F002 BETA.
Please enter command [ c | s | a | m | d | sf | sa | cn | u | r | l | x ] :- s
Please input fleet ID:- F001
Changed current fleet to F001.
Smart Delivery Fleet Management System (SDFMS)
c = create fleet, s = set current fleet, a = add driver, m = modify driver role,
d = delete driver, sf = show fleet, sa = display all fleets, cn = change fleet name,
u = undo, r = redo, l = list undo/redo, x = exit system
The current fleet is F001 ALPHA TWO.
Please enter command [ c | s | a | m | d | sf | sa | cn | u | r | l | x ] :- u
Command (Add driver, d101, Morgan, refrigerated-truck driver) is undone.
Smart Delivery Fleet Management System (SDFMS)
c = create fleet, s = set current fleet, a = add driver, m = modify driver role,
d = delete driver, sf = show fleet, sa = display all fleets, cn = change fleet name,
u = undo, r = redo, l = list undo/redo, x = exit system
The current fleet is F001 ALPHA TWO.
Please enter command [ c | s | a | m | d | sf | sa | cn | u | r | l | x ] :- sf
Delivery Van Fleet ALPHA TWO (F001)
Van Driver:
d001, Alex
Courier Driver: NIL
Smart Delivery Fleet Management System (SDFMS)
c = create fleet, s = set current fleet, a = add driver, m = modify driver role,
d = delete driver, sf = show fleet, sa = display all fleets, cn = change fleet name,
u = undo, r = redo, l = list undo/redo, x = exit system
The current fleet is F001 ALPHA TWO.
Please enter command [ c | s | a | m | d | sf | sa | cn | u | r | l | x ] :- s
Please input fleet ID:- F002
Changed current fleet to F002.
```

```
TC21  Undo updates the fleet the command was run on  (3/3)
```

```
Smart Delivery Fleet Management System (SDFMS)
c = create fleet, s = set current fleet, a = add driver, m = modify driver role,
d = delete driver, sf = show fleet, sa = display all fleets, cn = change fleet name,
u = undo, r = redo, l = list undo/redo, x = exit system
The current fleet is F002 BETA.
Please enter command [ c | s | a | m | d | sf | sa | cn | u | r | l | x ] :- sf
Refrigerated Truck Fleet BETA (F002)
Refrigerated-Truck Driver: NIL
Cold-Chain Delivery Driver: NIL
Smart Delivery Fleet Management System (SDFMS)
c = create fleet, s = set current fleet, a = add driver, m = modify driver role,
d = delete driver, sf = show fleet, sa = display all fleets, cn = change fleet name,
u = undo, r = redo, l = list undo/redo, x = exit system
The current fleet is F002 BETA.
Please enter command [ c | s | a | m | d | sf | sa | cn | u | r | l | x ] :- u
Command (Create refrigerated truck fleet, F002, BETA) is undone.
Smart Delivery Fleet Management System (SDFMS)
c = create fleet, s = set current fleet, a = add driver, m = modify driver role,
d = delete driver, sf = show fleet, sa = display all fleets, cn = change fleet name,
u = undo, r = redo, l = list undo/redo, x = exit system
The current fleet is F001 ALPHA TWO.
Please enter command [ c | s | a | m | d | sf | sa | cn | u | r | l | x ] :- u
Command (Change fleet name, F001, ALPHA TWO) is undone.
Smart Delivery Fleet Management System (SDFMS)
c = create fleet, s = set current fleet, a = add driver, m = modify driver role,
d = delete driver, sf = show fleet, sa = display all fleets, cn = change fleet name,
u = undo, r = redo, l = list undo/redo, x = exit system
The current fleet is F001 ALPHA.
Please enter command [ c | s | a | m | d | sf | sa | cn | u | r | l | x ] :- sf
Delivery Van Fleet ALPHA (F001)
Van Driver:
d001, Alex
Courier Driver: NIL
Smart Delivery Fleet Management System (SDFMS)
c = create fleet, s = set current fleet, a = add driver, m = modify driver role,
d = delete driver, sf = show fleet, sa = display all fleets, cn = change fleet name,
u = undo, r = redo, l = list undo/redo, x = exit system
The current fleet is F001 ALPHA.
Please enter command [ c | s | a | m | d | sf | sa | cn | u | r | l | x ] :- x
Thank you for using SDFMS. System terminated.
```

TC22 Undo and redo of create fleet

Steps. Create a fleet, add a driver, undo both, display all, redo both, show.

Expected result. After the two undos there is no fleet record. After the two redos the fleet and the driver are back.

Result. Pass. The capture below is the program output.

TC22 Undo and redo of create fleet (1/2)

```
Smart Delivery Fleet Management System (SDFMS)
c = create fleet, s = set current fleet, a = add driver, m = modify driver role,
d = delete driver, sf = show fleet, sa = display all fleets, cn = change fleet name,
u = undo, r = redo, l = list undo/redo, x = exit system
Please enter command [ c | s | a | m | d | sf | sa | cn | u | r | l | x ] :- c
Enter fleet type (v = delivery van | r = refrigerated truck) :- v
Fleet ID:- F001
Fleet Name:- ALPHA
Delivery van fleet is created.
Current fleet is changed to F001.
Smart Delivery Fleet Management System (SDFMS)
c = create fleet, s = set current fleet, a = add driver, m = modify driver role,
d = delete driver, sf = show fleet, sa = display all fleets, cn = change fleet name,
u = undo, r = redo, l = list undo/redo, x = exit system
The current fleet is F001 ALPHA.
Please enter command [ c | s | a | m | d | sf | sa | cn | u | r | l | x ] :- a
Please input driver information (id, name):- d1, Ann
Role (1 = van driver | 2 = courier driver) :- 1
Driver is added.
Smart Delivery Fleet Management System (SDFMS)
c = create fleet, s = set current fleet, a = add driver, m = modify driver role,
d = delete driver, sf = show fleet, sa = display all fleets, cn = change fleet name,
u = undo, r = redo, l = list undo/redo, x = exit system
The current fleet is F001 ALPHA.
Please enter command [ c | s | a | m | d | sf | sa | cn | u | r | l | x ] :- u
Command (Add driver, d1, Ann, van driver) is undone.
Smart Delivery Fleet Management System (SDFMS)
c = create fleet, s = set current fleet, a = add driver, m = modify driver role,
d = delete driver, sf = show fleet, sa = display all fleets, cn = change fleet name,
u = undo, r = redo, l = list undo/redo, x = exit system
The current fleet is F001 ALPHA.
Please enter command [ c | s | a | m | d | sf | sa | cn | u | r | l | x ] :- u
Command (Create delivery van fleet, F001, ALPHA) is undone.
Smart Delivery Fleet Management System (SDFMS)
c = create fleet, s = set current fleet, a = add driver, m = modify driver role,
d = delete driver, sf = show fleet, sa = display all fleets, cn = change fleet name,
u = undo, r = redo, l = list undo/redo, x = exit system
Please enter command [ c | s | a | m | d | sf | sa | cn | u | r | l | x ] :- sa
No fleet record!
Smart Delivery Fleet Management System (SDFMS)
```

TC22 Undo and redo of create fleet (2/2)

```
c = create fleet, s = set current fleet, a = add driver, m = modify driver role,
d = delete driver, sf = show fleet, sa = display all fleets, cn = change fleet name,
u = undo, r = redo, l = list undo/redo, x = exit system
Please enter command [ c | s | a | m | d | sf | sa | cn | u | r | l | x ] :- r
Command (Create delivery van fleet, F001, ALPHA) is redone.
Smart Delivery Fleet Management System (SDFMS)
c = create fleet, s = set current fleet, a = add driver, m = modify driver role,
d = delete driver, sf = show fleet, sa = display all fleets, cn = change fleet name,
u = undo, r = redo, l = list undo/redo, x = exit system
The current fleet is F001 ALPHA.
Please enter command [ c | s | a | m | d | sf | sa | cn | u | r | l | x ] :- r
Command (Add driver, d1, Ann, van driver) is redone.
Smart Delivery Fleet Management System (SDFMS)
c = create fleet, s = set current fleet, a = add driver, m = modify driver role,
d = delete driver, sf = show fleet, sa = display all fleets, cn = change fleet name,
u = undo, r = redo, l = list undo/redo, x = exit system
The current fleet is F001 ALPHA.
Please enter command [ c | s | a | m | d | sf | sa | cn | u | r | l | x ] :- sf
Delivery Van Fleet ALPHA (F001)
Van Driver:
d1, Ann
Courier Driver: NIL
Smart Delivery Fleet Management System (SDFMS)
c = create fleet, s = set current fleet, a = add driver, m = modify driver role,
d = delete driver, sf = show fleet, sa = display all fleets, cn = change fleet name,
u = undo, r = redo, l = list undo/redo, x = exit system
The current fleet is F001 ALPHA.
Please enter command [ c | s | a | m | d | sf | sa | cn | u | r | l | x ] :- x
Thank you for using SDFMS. System terminated.
```

TC23 Empty fleet name

Steps. cn with an empty name, then l.

Expected result. The name is rejected. The undo list still contains only the create.

Result. Pass. The capture below is the program output.

```
TC23  Empty fleet name is rejected and is not recorded

Smart Delivery Fleet Management System (SDFMS)
c = create fleet, s = set current fleet, a = add driver, m = modify driver role,
d = delete driver, sf = show fleet, sa = display all fleets, cn = change fleet name,
u = undo, r = redo, l = list undo/redo, x = exit system
Please enter command [ c | s | a | m | d | sf | sa | cn | u | r | l | x ] :- c
Enter fleet type (v = delivery van | r = refrigerated truck) :- v
Fleet ID:- F001
Fleet Name:- ALPHA
Delivery van fleet is created.
Current fleet is changed to F001.
Smart Delivery Fleet Management System (SDFMS)
c = create fleet, s = set current fleet, a = add driver, m = modify driver role,
d = delete driver, sf = show fleet, sa = display all fleets, cn = change fleet name,
u = undo, r = redo, l = list undo/redo, x = exit system
The current fleet is F001 ALPHA.
Please enter command [ c | s | a | m | d | sf | sa | cn | u | r | l | x ] :- cn
Please input new name of the current fleet:-
Fleet name cannot be empty!
Smart Delivery Fleet Management System (SDFMS)
c = create fleet, s = set current fleet, a = add driver, m = modify driver role,
d = delete driver, sf = show fleet, sa = display all fleets, cn = change fleet name,
u = undo, r = redo, l = list undo/redo, x = exit system
The current fleet is F001 ALPHA.
Please enter command [ c | s | a | m | d | sf | sa | cn | u | r | l | x ] :- l
Undo List
Create delivery van fleet, F001, ALPHA
-- End of undo list --
Redo List
-- End of redo list --
Smart Delivery Fleet Management System (SDFMS)
c = create fleet, s = set current fleet, a = add driver, m = modify driver role,
d = delete driver, sf = show fleet, sa = display all fleets, cn = change fleet name,
u = undo, r = redo, l = list undo/redo, x = exit system
The current fleet is F001 ALPHA.
Please enter command [ c | s | a | m | d | sf | sa | cn | u | r | l | x ] :- x
Thank you for using SDFMS. System terminated.
```

TC24 Failed commands are not recorded

Steps. Unknown fleet, unknown driver, bad driver text, invalid command, then l.

Expected result. The undo list contains only the successful create. The redo list is empty.

Result. Pass. The capture below is the program output.

TC24 Failed commands are not added to the undo list (1/2)

```
Smart Delivery Fleet Management System (SDFMS)
c = create fleet, s = set current fleet, a = add driver, m = modify driver role,
d = delete driver, sf = show fleet, sa = display all fleets, cn = change fleet name,
u = undo, r = redo, l = list undo/redo, x = exit system
Please enter command [ c | s | a | m | d | sf | sa | cn | u | r | l | x ] :- c
Enter fleet type (v = delivery van | r = refrigerated truck) :- v
Fleet ID:- F001
Fleet Name:- ALPHA
Delivery van fleet is created.
Current fleet is changed to F001.
Smart Delivery Fleet Management System (SDFMS)
c = create fleet, s = set current fleet, a = add driver, m = modify driver role,
d = delete driver, sf = show fleet, sa = display all fleets, cn = change fleet name,
u = undo, r = redo, l = list undo/redo, x = exit system
The current fleet is F001 ALPHA.
Please enter command [ c | s | a | m | d | sf | sa | cn | u | r | l | x ] :- s
Please input fleet ID:- NOPE
Fleet NOPE is not found!
Smart Delivery Fleet Management System (SDFMS)
c = create fleet, s = set current fleet, a = add driver, m = modify driver role,
d = delete driver, sf = show fleet, sa = display all fleets, cn = change fleet name,
u = undo, r = redo, l = list undo/redo, x = exit system
The current fleet is F001 ALPHA.
Please enter command [ c | s | a | m | d | sf | sa | cn | u | r | l | x ] :- d
Please input driver ID:- NOPE
Driver NOPE is not found in the current fleet!
Smart Delivery Fleet Management System (SDFMS)
c = create fleet, s = set current fleet, a = add driver, m = modify driver role,
d = delete driver, sf = show fleet, sa = display all fleets, cn = change fleet name,
u = undo, r = redo, l = list undo/redo, x = exit system
The current fleet is F001 ALPHA.
Please enter command [ c | s | a | m | d | sf | sa | cn | u | r | l | x ] :- m
Please input driver ID:- NOPE
Driver NOPE is not found in the current fleet!
Smart Delivery Fleet Management System (SDFMS)
c = create fleet, s = set current fleet, a = add driver, m = modify driver role,
d = delete driver, sf = show fleet, sa = display all fleets, cn = change fleet name,
u = undo, r = redo, l = list undo/redo, x = exit system
The current fleet is F001 ALPHA.
Please enter command [ c | s | a | m | d | sf | sa | cn | u | r | l | x ] :- a
```

TC24 Failed commands are not added to the undo list (2/2)

```
Please input driver information (id, name):- bad
Invalid driver information!
Smart Delivery Fleet Management System (SDFMS)
c = create fleet, s = set current fleet, a = add driver, m = modify driver role,
d = delete driver, sf = show fleet, sa = display all fleets, cn = change fleet name,
u = undo, r = redo, l = list undo/redo, x = exit system
The current fleet is F001 ALPHA.
Please enter command [ c | s | a | m | d | sf | sa | cn | u | r | l | x ] :- q
Invalid command!
Smart Delivery Fleet Management System (SDFMS)
c = create fleet, s = set current fleet, a = add driver, m = modify driver role,
d = delete driver, sf = show fleet, sa = display all fleets, cn = change fleet name,
u = undo, r = redo, l = list undo/redo, x = exit system
The current fleet is F001 ALPHA.
Please enter command [ c | s | a | m | d | sf | sa | cn | u | r | l | x ] :- l
Undo List
Create delivery van fleet, F001, ALPHA
-- End of undo list --
Redo List
-- End of redo list --
Smart Delivery Fleet Management System (SDFMS)
c = create fleet, s = set current fleet, a = add driver, m = modify driver role,
d = delete driver, sf = show fleet, sa = display all fleets, cn = change fleet name,
u = undo, r = redo, l = list undo/redo, x = exit system
The current fleet is F001 ALPHA.
Please enter command [ c | s | a | m | d | sf | sa | cn | u | r | l | x ] :- x
Thank you for using SDFMS. System terminated.
```

TC25 Memento undo and redo on a refrigerated truck

Steps. Add d1 as cold-chain delivery driver, change the role to refrigerated-truck driver, undo, show, redo, show.

Expected result. Undo puts d1 back under Cold-Chain Delivery Driver. Redo puts d1 under Refrigerated-Truck Driver.

Result. Pass. The capture below is the program output.

```
TC25  Memento undo and redo of a refrigerated-truck role  (1/2)

Smart Delivery Fleet Management System (SDFMS)
c = create fleet, s = set current fleet, a = add driver, m = modify driver role,
d = delete driver, sf = show fleet, sa = display all fleets, cn = change fleet name,
u = undo, r = redo, l = list undo/redo, x = exit system
Please enter command [ c | s | a | m | d | sf | sa | cn | u | r | l | x ] :- c
Enter fleet type (v = delivery van | r = refrigerated truck) :- r
Fleet ID:- F102
Fleet Name:- FROST
Refrigerated truck fleet is created.
Current fleet is changed to F102.
Smart Delivery Fleet Management System (SDFMS)
c = create fleet, s = set current fleet, a = add driver, m = modify driver role,
d = delete driver, sf = show fleet, sa = display all fleets, cn = change fleet name,
u = undo, r = redo, l = list undo/redo, x = exit system
The current fleet is F102 FROST.
Please enter command [ c | s | a | m | d | sf | sa | cn | u | r | l | x ] :- a
Please input driver information (id, name):- d1, Mo
Role (1 = refrigerated-truck driver | 2 = cold-chain delivery driver) :- 2
Driver is added.
Smart Delivery Fleet Management System (SDFMS)
c = create fleet, s = set current fleet, a = add driver, m = modify driver role,
d = delete driver, sf = show fleet, sa = display all fleets, cn = change fleet name,
u = undo, r = redo, l = list undo/redo, x = exit system
The current fleet is F102 FROST.
Please enter command [ c | s | a | m | d | sf | sa | cn | u | r | l | x ] :- m
Please input driver ID:- d1
Role (1 = refrigerated-truck driver | 2 = cold-chain delivery driver) :- 1
Driver role is updated.
Smart Delivery Fleet Management System (SDFMS)
c = create fleet, s = set current fleet, a = add driver, m = modify driver role,
d = delete driver, sf = show fleet, sa = display all fleets, cn = change fleet name,
u = undo, r = redo, l = list undo/redo, x = exit system
The current fleet is F102 FROST.
Please enter command [ c | s | a | m | d | sf | sa | cn | u | r | l | x ] :- l
Undo List
Create refrigerated truck fleet, F102, FROST
Add driver, d1, Mo, cold-chain delivery driver
Modify driver role, d1, refrigerated-truck driver
-- End of undo list --
Redo List
```

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```
-- End of redo list --
Smart Delivery Fleet Management System (SDFMS)
c = create fleet, s = set current fleet, a = add driver, m = modify driver role,
d = delete driver, sf = show fleet, sa = display all fleets, cn = change fleet name,
u = undo, r = redo, l = list undo/redo, x = exit system
The current fleet is F102 FROST.
Please enter command [ c | s | a | m | d | sf | sa | cn | u | r | l | x ] :- u
Command (Modify driver role, d1, refrigerated-truck driver) is undone.
Smart Delivery Fleet Management System (SDFMS)
c = create fleet, s = set current fleet, a = add driver, m = modify driver role,
d = delete driver, sf = show fleet, sa = display all fleets, cn = change fleet name,
u = undo, r = redo, l = list undo/redo, x = exit system
The current fleet is F102 FROST.
Please enter command [ c | s | a | m | d | sf | sa | cn | u | r | l | x ] :- sf
Refrigerated Truck Fleet FROST (F102)
Refrigerated-Truck Driver: NIL
Cold-Chain Delivery Driver:
d1, Mo
Smart Delivery Fleet Management System (SDFMS)
c = create fleet, s = set current fleet, a = add driver, m = modify driver role,
d = delete driver, sf = show fleet, sa = display all fleets, cn = change fleet name,
u = undo, r = redo, l = list undo/redo, x = exit system
The current fleet is F102 FROST.
Please enter command [ c | s | a | m | d | sf | sa | cn | u | r | l | x ] :- r
Command (Modify driver role, d1, refrigerated-truck driver) is redone.
Smart Delivery Fleet Management System (SDFMS)
c = create fleet, s = set current fleet, a = add driver, m = modify driver role,
d = delete driver, sf = show fleet, sa = display all fleets, cn = change fleet name,
u = undo, r = redo, l = list undo/redo, x = exit system
The current fleet is F102 FROST.
Please enter command [ c | s | a | m | d | sf | sa | cn | u | r | l | x ] :- sf
Refrigerated Truck Fleet FROST (F102)
Refrigerated-Truck Driver:
d1, Mo
Cold-Chain Delivery Driver: NIL
Smart Delivery Fleet Management System (SDFMS)
c = create fleet, s = set current fleet, a = add driver, m = modify driver role,
d = delete driver, sf = show fleet, sa = display all fleets, cn = change fleet name,
u = undo, r = redo, l = list undo/redo, x = exit system
The current fleet is F102 FROST.
Please enter command [ c | s | a | m | d | sf | sa | cn | u | r | l | x ] :- x
Thank you for using SDFMS. System terminated.
```

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```
Please enter command [ c | s | a | m | d | sf | sa | cn | u | r | l | x ] :- x
Thank you for using SDFMS. System terminated.
```

6. Source code

The source is the `.java` files in the same folder as `run.bat`. Each class has a file comment that states its pattern role, and public methods that are not obvious from the class diagram have a short comment. There are 25 source files. `javac *.java` compiles them because they are all in the default package.

File	Role
Main.java	Client and menu loop
Command.java	Command interface
UndoableCommand.java	Undoable command interface
Caretaker.java	Undo stack and redo stack

CommandFactory.java	Creates command objects
CreateFleetCommand.java	Create fleet
SetCurrentFleetCommand.java	Set current fleet
AddDriverCommand.java	Add driver
ModifyDriverRoleCommand.java	Modify role, keeps role mementos
DeleteDriverCommand.java	Delete driver
ShowFleetCommand.java	Show current fleet
DisplayAllFleetsCommand.java	Display all fleets
ChangeFleetNameCommand.java	Rename fleet, keeps name mementos
UndoCommand.java	Undo
RedoCommand.java	Redo
ListUndoRedoCommand.java	Show undo and redo lists
VehicleFleetFactory.java	Creates fleet objects
DriverFactory.java	Creates driver objects
FleetSystem.java	Receiver: all fleets and the current fleet
VehicleFleet.java	Abstract fleet from the class diagram
DeliveryVanFleet.java	Delivery van fleet
RefrigeratedTruckFleet.java	Refrigerated truck fleet
Driver.java	Driver from the class diagram
DriverRoleMemento.java	Memento of a driver's role
FleetNameMemento.java	Memento of a fleet name
run.bat	Compile and run on Windows

Output is produced by walking the fleets and drivers that were entered. No sample name is printed unless the user typed it.